HotelAndStars

1. Objective.

Build an Entity-Relationship Model (ERM) and, based on it, a Dimensional Model for the manager of the fictional company HotelAndStars, considering the following demands:

- The manager wants to know:
  - How many accommodations and of what type are sold per day;
  - What is the daily revenue;
  - Where the customers are from: country, state, and city;
  - How holidays and school vacations influence accommodations.
- Does not need to know:
  - Who stayed.

2. Entity-Relationship Model (ERM) - SQL Server Management Studio (SSMS).

ERM created:

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3. Dimensional Model - SQL Server Management Studio (SSMS).

1. To create the analytical database structure model (OLAP) considering the transactional database (OLTP), the subject to be analyzed (the fact) and the dimensions are the various perspectives from which the fact is to be analyzed (the dimensions), using the Star Schema model (the most common model for Business Intelligence), the following transformations were made:

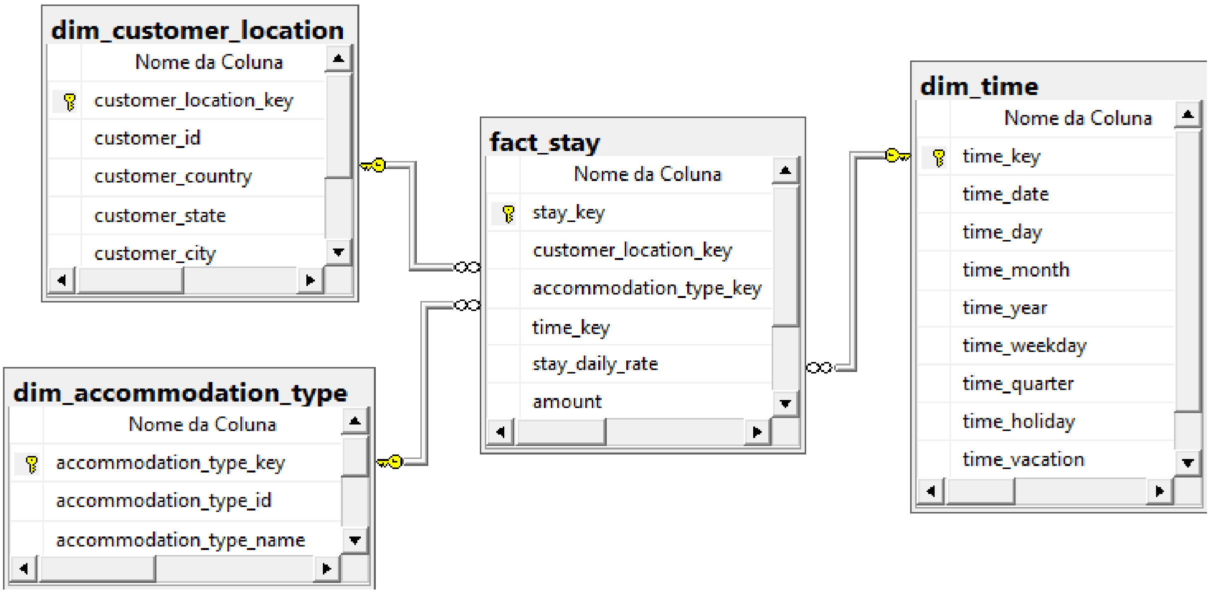
Dimension	Notes	Attributes	Source Table
dim_accomodation_type (type of accommodation sold)	Retrieves the ID of the accommodation type and its name from the	accommodation_type_key (SK)	-

	entity-relationship model.	accommodation_type_id	accommodation_type
		accommodation_type_name	accommodation_type
dim_customer_location (where are the customers from?)	Retrieves the customer ID and its location attributes related to country, state, and city from the entity-relationship model.	customer_location_key (SK)	-
		customer_id	customer
		customer_country	customer
		customer_state	customer
		customer_city	customer
dim_time (dates)	-	time_key (SK)	-
		time_date	-
		time_day	-
		time_month	-
		time_year	-
		time_weekday	-
		time_quarter	-
		time_holiday	-
		time_vacation	-

Fact	**Consolidated table**	**Granularity**	**Notes**	**Attributes**	**Source table**
fact_stay	stay	Information on accommodations by day.	Retrieves the daily rate for each lodging stay from the entity-relationship model.	stay_key (SK)	-
				customer_location_key (FK)	dim_customer_location
				accommodation_type_key (FK)	dim_accomodation_type
				time_key (FK)	dim_time
				stay_daily_rate	stay
				amount (The default value is 1 for all records, aiming for	-

				aggregation by summation)	

4. Dimensional Model created:



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